PROJETO TMBA - Google Play Store App

Projeto desenvolvido por:

- Débora Jesus, nº2250064
- João Vasques, nº2250055
- Rui Filipe Parada, nº2211025

LLM: DistilGPT2

1. Imports and Data Upload

```
# Instalação de bibliotecas adicionais
!pip install -q transformers datasets scikit-learn
!pip install datasets
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.29.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
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Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (20
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.11.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.1
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.28.1)
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2026.4.22)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2026.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0-
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=202
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (4.13.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (8.3.3)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub>=0.24.0->datasets
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub>=0.24.0->datase
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub>=0.24
```

```
# Imports principais
from google.colab import drive
from google.colab import files
import pandas as pd
import numpy as np
import io
import torch

# Pré-processamento e métricas
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    cohen_kappa_score,
    confusion_matrix
)

# Hugging Face - tokenização, modelo e treino
from transformers import (
    AutoTokenizer,
    GPT2ForSequenceClassification,
    TrainingArguments,
    Trainer
)
```

```
# Conversão para dataset compatível com Hugging Face
from datasets import Dataset
```

```
# Upload do dataset (em csv)

uploaded = files.upload()

review_df = pd.read_csv(io.BytesIO(uploaded['google_play_store_apps.csv']), sep=',')
```

Escolher Ficheiros Nenhum ficheiro selecionado Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving google_play_store_apps.csv to google_play_store_apps.csv

```
review_df.head()
```

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.00	0.533333
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.25	0.288462
2	10 Best Foods for You	Works great especially going grocery store	Positive	0.40	0.875000
3	10 Best Foods for You	Best idea us	Positive	1.00	0.300000
4	10 Best Foods for You	Best way	Positive	1.00	0.300000

2. Pré-processamento

```
# Remover reviews com sentimento neutro
review_df = review_df[review_df['Sentiment'] != 'Neutral']

# Visualizar as primeiras linhas
display(review_df.head(10))
```

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.000000	0.533333
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.250000	0.288462
2	10 Best Foods for You	Works great especially going grocery store	Positive	0.400000	0.875000
3	10 Best Foods for You	Best idea us	Positive	1.000000	0.300000
4	10 Best Foods for You	Best way	Positive	1.000000	0.300000
5	10 Best Foods for You	Amazing	Positive	0.600000	0.900000
8	10 Best Foods for You	good you.	Positive	0.700000	0.600000
9	10 Best Foods for You	Useful information The amount spelling errors ...	Positive	0.200000	0.100000
10	10 Best Foods for You	Thank you! Great app!! Add arthritis, eyes, im...	Positive	0.750000	0.875000
11	10 Best Foods for You	Greatest ever Completely awesome maintain heal...	Positive	0.992188	0.866667

3. Fine Tuning

3.1. Codificação dos Rótulos

```
# Codificar sentimentos em valores numéricos
le = LabelEncoder()
review_df["label"] = le.fit_transform(review_df["Sentiment"])

# Mostrar primeiras linhas
display(review_df.head(10))
```

	App	Translated_Review	Sentiment	Sentiment_Polarity	Sentiment_Subjectivity	label
0	10 Best Foods for You	I like eat delicious food. That's I'm cooking ...	Positive	1.000000	0.533333	1
1	10 Best Foods for You	This help eating healthy exercise regular basis	Positive	0.250000	0.288462	1
2	10 Best Foods for You	Works great especially going grocery store	Positive	0.400000	0.875000	1
3	10 Best Foods for You	Best idea us	Positive	1.000000	0.300000	1
4	10 Best Foods for You	Best way	Positive	1.000000	0.300000	1
5	10 Best Foods for You	Amazing	Positive	0.600000	0.900000	1
8	10 Best Foods for You	good you.	Positive	0.700000	0.600000	1
9	10 Best Foods for You	Useful information The amount spelling errors ...	Positive	0.200000	0.100000	1
10	10 Best Foods for You	Thank you! Great app!! Add arthritis, eyes, im...	Positive	0.750000	0.875000	1
11	10 Best Foods for You	Greatest ever Completely awesome maintain heal...	Positive	0.992188	0.866667	1

3.2. Separação dos Dados em Treino e Teste

```
# Separar dados de treino (80%) e teste (20%)
train_df = review_df.sample(frac=0.8, random_state=42)
test_df = review_df.drop(train_df.index)
```

```
# Mostrar dimensões dos conjuntos
print("train_df:", train_df.shape)
print("→", train_df.shape[0], "linhas (documentos de treino) e", train_df.shape[1], "colunas (atributos seleccionadas)")

print("\ntest_df:", test_df.shape)
print("→", test_df.shape[0], "linhas (documentos de teste) e", test_df.shape[1], "colunas (atributos seleccionadas)")
```

```
train_df: (25815, 6)
→ 25815 linhas (documentos de treino) e 6 colunas (atributos seleccionadas)

test_df: (6454, 6)
→ 6454 linhas (documentos de teste) e 6 colunas (atributos seleccionadas)
```

3.3. Conversão para Dataset Hugging Face

```
# Converter DataFrames para Dataset Hugging Face
train_dataset = Dataset.from_pandas(train_df[["Translated_Review", "label"]])
test_dataset = Dataset.from_pandas(test_df[["Translated_Review", "label"]])
```

```
# Mostrar o conteúdo
display(train_dataset)
display(test_dataset)
```

```
Dataset({
    features: ['Translated_Review', 'label', '__index_level_0__'],
    num_rows: 25815
})
Dataset({
    features: ['Translated_Review', 'label', '__index_level_0__'],
    num_rows: 6454
})
```

3.4. Tokenização dos Conjuntos de Treino e Teste

```
# Carregar tokenizador do DistilGPT2
tokenizer = AutoTokenizer.from_pretrained("distilgpt2")

# Definir token de padding
tokenizer.pad_token = tokenizer.eos_token

# Tokenizar treino
train_encodings = tokenizer(
    list(train_dataset["Translated_Review"]),
    truncation=True,
    padding=True,
    max_length=512
)

# Tokenizar teste
test_encodings = tokenizer(
    list(test_dataset["Translated_Review"]),
    truncation=True,
    padding=True,
    max_length=512
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Co
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
```

```
# Adicionar tokens ao treino
train_dataset = train_dataset.add_column("input_ids", train_encodings["input_ids"])
train_dataset = train_dataset.add_column("attention_mask", train_encodings["attention_mask"])

# Adicionar tokens ao teste
test_dataset = test_dataset.add_column("input_ids", test_encodings["input_ids"])
test_dataset = test_dataset.add_column("attention_mask", test_encodings["attention_mask"])
```

3.5. Carregamento do Modelo e Definição dos Parâmetros de Treino

```
# Converter datasets para formato PyTorch
train_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])
test_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])

# Carregar modelo para classificação binária
model = GPT2ForSequenceClassification.from_pretrained(
    "distilgpt2",
    num_labels=2
)

# Informar o modelo sobre o pad token
model.config.pad_token_id = tokenizer.pad_token_id

# Definir parâmetros de treino
training_args = TrainingArguments(
    output_dir="./output",
    per_device_train_batch_size=4,
    per_device_eval_batch_size=4,
    num_train_epochs=2,
    report_to="none"
)
```

```
Loading weights: 100%                                                    76/76 [00:00<00:00, 417.75it/s, Materializing param=transformer.wte.weight]
GPT2ForSequenceClassification LOAD REPORT from: distilgpt2
Key                                        | Status     |
-------------------------------------------+------------+-
transformer.h.{0, 1, 2, 3, 4, 5}.attn.bias | UNEXPECTED |
score.weight                               | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

3.6. Definição da Métrica de Avaliação

```
def calcular_metricas(eval_pred):

    logits, labels = eval_pred

    preds = np.argmax(logits, axis=1)

    acc = accuracy_score(labels, preds)
    prec = precision_score(labels, preds, average='weighted')
    rec = recall_score(labels, preds, average='weighted')
    f1 = f1_score(labels, preds, average='weighted')
    kappa = cohen_kappa_score(labels, preds)

    cm = confusion_matrix(labels, preds)

    tn, fp, fn, tp = cm.ravel()

    taxa_erro = (fp + fn) / (tp + tn + fp + fn)

    return {
        "accuracy": acc,
        "precision": prec,
        "recall": rec,
        "f1": f1,
        "kappa": kappa,
        "taxa_erro": taxa_erro
    }
```

3.7. Treino e Avaliação do Modelo

```
# Configurar treino
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=calcular_metricas
)

# Treinar modelo
trainer.train()
```

[7501/12908 39:46 < 28:40, 3.14 it/s, Epoch 1.16/2]

Step	Training Loss
500	0.520167
1000	0.458586
1500	0.396575
2000	0.380799
2500	0.390117
3000	0.393662
3500	0.380713
4000	0.380089
4500	0.322493
5000	0.306549
5500	0.287903

```
# Avaliar modelo
resultados = trainer.evaluate()

# Converter resultados para DataFrame
df_resultados = pd.DataFrame([{
    "Model": "DistilGPT2",
    "Acc": resultados["eval_accuracy"],
    "Prec": resultados["eval_precision"],
    "Rec": resultados["eval_recall"],
    "F1": resultados["eval_f1"],
    "Kappa": resultados["eval_kappa"],
    "Taxa_Erro": resultados["eval_taxa_erro"]
}])

# Mostrar tabela
print(df_resultados)

# Exportar CSV
df_resultados.to_csv("DistilGPT2_results.csv", index=False)
```

Writing model shards: 100% [1614/1614 00:54] 1/1 [00:17<00:00, 17.67s/it]

```
      Model       Acc      Prec       Rec        F1     Kappa  Taxa_Erro
0  DistilGPT2  0.951348  0.951874  0.951348  0.951556  0.873424   0.048652
```

Writing model shards: 100% 1/1 [00:26<00:00, 26.18s/it]

Writing model shards: 100% 1/1 [00:12<00:00, 12.02s/it]

Writing model shards: 100% 1/1 [00:28<00:00, 28.19s/it]

Writing model shards: 100% 1/1 [00:16<00:00, 16.48s/it]

4. Predição de Novo Comentário

```
### --- NOVOS COMENTÁRIOS: --- ###

uploaded = files.upload()

novos_comentarios = pd.read_csv(io.BytesIO(uploaded['Comentarios.csv']), sep=';')
```

Escolher Ficheiros Nenhum ficheiro selecionado Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving Comentarios.csv to Comentarios.csv

2000	0.380799
2500	0.390117
3000	0.393662
3500	0.380713
4000	0.380089
4500	0.322493
5000	0.306549
5500	0.287903
6000	0.252125
6500	0.234393
7000	0.198414
7500	0.198414
8000	0.221709
8500	0.234393
9000	0.194372

4.1. Pré-processamento

```
# Remover reviews com sentimento neutro
novos_comentarios = novos_comentarios[novos_comentarios['Sentiment'] != 'Neutral']

# Visualizar as primeiras linhas
display(novos_comentarios.head(10))
```

	Comentarios_Novos	Sentiment
0	This app is a true paragon.	Positive
1	The interface remains totally execrable.	Negative
3	Simply effulgent and highly efficient!	Positive
4	Navigation is inimical to usage.	Negative

4.2. Avaliação de Novos Comentários

```
# Extrair textos
textos = novos_comentarios["Comentarios_Novos"].astype(str).tolist()

# Tokenizar em batch
tokens = tokenizer(
    textos,
    truncation=True,
    padding=True,
    max_length=128,
    return_tensors="pt"
)
```

```
# Garantir que o modelo está na GPU (se existir)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)

# Enviar tokens para o mesmo device
tokens = {k: v.to(device) for k, v in tokens.items()}

# Colocar modelo em modo avaliação
model.eval()

# Inferência
with torch.no_grad():
    outputs = model(**tokens)

logits = outputs.logits

# Previsões
classe_prevista = np.argmax(logits.cpu().numpy(), axis=1)

# Converter para labels originais
sentimento = le.inverse_transform(classe_prevista)

# Guardar no dataframe
novos_comentarios["Sentimento_previsto"] = sentimento

# Mostrar resultados
print(novos_comentarios[["Comentarios_Novos", "Sentiment", "Sentimento_previsto"]])
```

```
                        Comentarios_Novos Sentiment Sentimento_previsto
0              This app is a true paragon.  Positive            Positive
1  The interface remains totally execrable.  Negative            Negative
3    Simply effulgent and highly efficient!  Positive            Positive
4          Navigation is inimical to usage.  Negative            Negative
```